



3.22 AVIATION TECHNOLOGY (450)

Aviation Technology Paper

SECTION A (10 marks)

Answer all the questions in this section

- Outline **three** roles of a ground controller in airport operations. (3 marks)
- a** Explain **two** methods of controlling movement of tools in an aircraft hangar. (2 marks)
- b** State **four** methods of alerting personnel in the event of fire outbreak. (4 marks)
- c** State **four** reasons of using timber in aircraft construction. (4 marks)
- Differentiate between the following:
- (a) airport and airfield; (2 marks)
- b** runway and taxiway. (2 marks)
- a** State **four** causes of flow change from laminar to turbulent on an aircraft wing. (4 marks)
- b** Use a labelled sketch to show how lift is generated on an aerofoil. (2 marks)
- Explain each of the following terms as applied to aircraft structure
- a** fairing (1 mark)
- b** monocoque (1 mark)
- c** rigging position (1 mark)
- d** winglet (1 mark)
- Use a labelled sketch to show the forces acting on an aircraft propeller blade during generation of thrust. (2 marks)
- a** Explain **two** methods of preventing failure in an aircraft hydraulic system. (2 marks)
- (b) State the meaning and the function of each of the following aircraft flight systems:
- i** RMI (1½ marks)
- (ii) DME (1½ marks)
- i** ILS (1½ marks)



□□ Explain **four** advantages which make aluminium alloys best suited for aircraft fabrication. **3** marks

□□□ Figure 1 shows a bracket drawn in isometric projection.

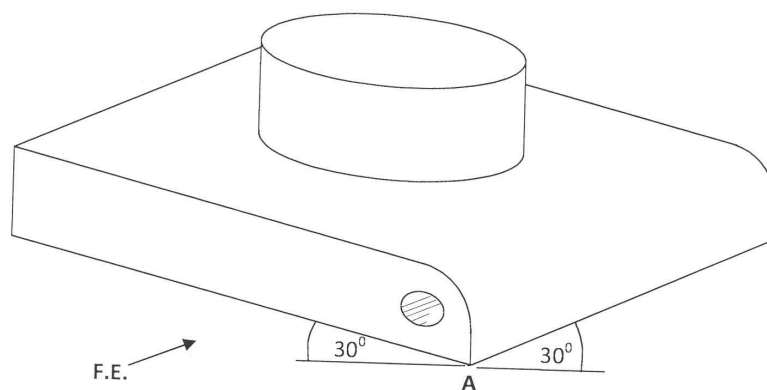


Figure 1

□ Sketch in good proportion the orthographic views of the bracket in third angle projection. **4 1/2** marks



SECTION B [10] Marks

Answer any four questions from this section

Candidates are advised to spend not more than 10 minutes on question [10]

- [10] Figure 2 shows the three orthographic views of an aircraft engine bracket drawn in first angle projection

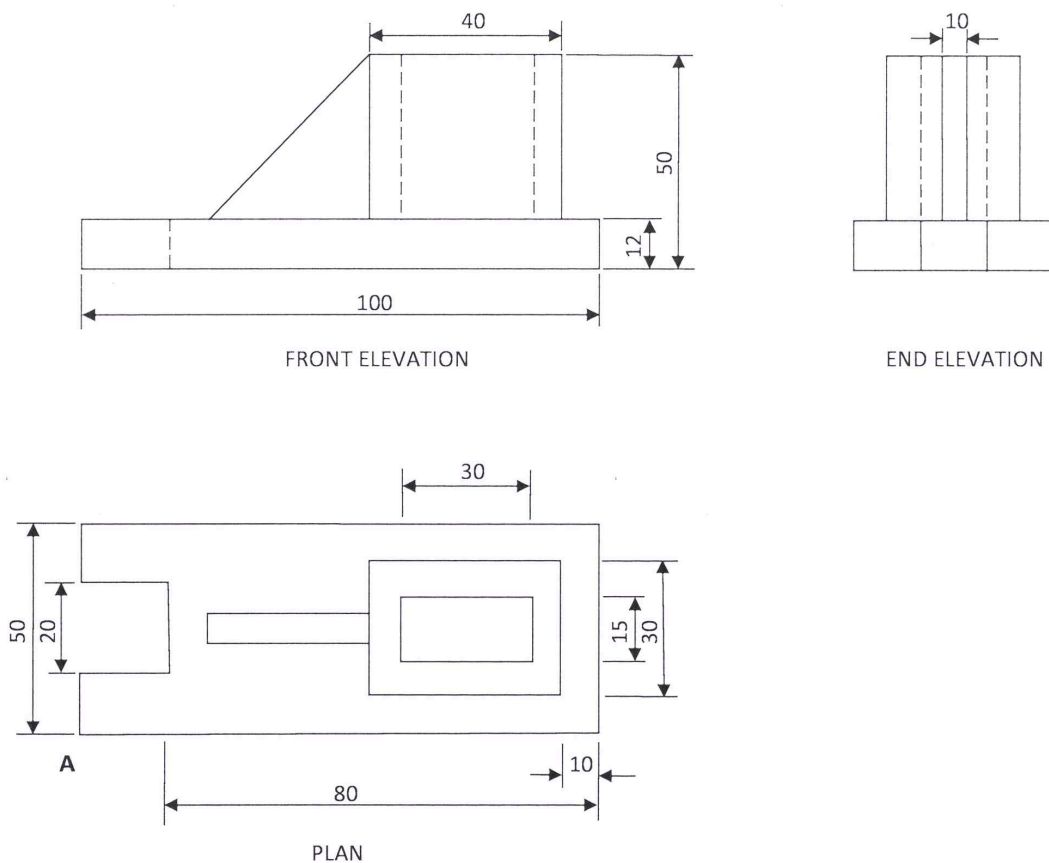


Figure 2

- [10] On the isometric grid paper provided on page [10] draw the isometric view of the bracket taking A as the lowest point

[10] Marks

- [10] With the aid of pressure volume diagrams describe the operation of the following aircraft engines

- | | | | | | | | | | | |
|------|-------------|------|------|------|------|------|------|------|------|------|
| [10] | aeropiston | [10] | [10] | [10] | [10] | [10] | [10] | [10] | [10] | [10] |
| [10] | gas turbine | [10] | [10] | [10] | [10] | [10] | [10] | [10] | [10] | [10] |



Q1. With the aid of a labelled schematic diagram explain the operation of a typical pneumatic emergency system. (4 marks)

Q2. State four reasons why a pneumatic system is preferred to a hydraulic system. (4 marks)

Q3. Outline four functions of aircraft control tabs in flight. (4 marks)

Q4. With the aid of a labelled diagram explain the operation of a spring servo tab in pitching mode. (4 marks)

Q5. State four advantages of flying an aircraft in the stratosphere layer. (4 marks)

Q6. Explain the meaning of each of the following markings on a runway.

Q6.1 yellow on black (1 mark)

Q6.2 black on yellow (1 mark)

Q6.3 stop and yield (1 mark)

Q6.4 white on red (1 mark)

(c) An aircraft with wing span of 220m and a chord of 25m is flying at 850 KPH at sea level where the density is 1.225 kg/m^3 . If the lift and drag coefficients are 0.545 and 0.025 respectively determine

Q6.5 lift (1 mark)

Q6.6 drag (1 mark)